

IT PROJECT MANAGEMENT

TOPIC 12

EXECUTING PROJECTS

LEARNING OBJECTIVES

a. List the **processes** and **outputs** of project execution when using a predictive approach to project management

b. Discuss what is involved in **directing** and **managing** project work and managing **project knowledge** as part of project integration management, including the importance of producing promised deliverables, implementing solutions to issues, evaluating work performance data, preparing issue logs, requesting changes to a project, and managing lessons learned

c. Explain the importance of **creating quality reports**, test and evaluation documents, and change requests as part of quality assurance

d. Describe the **executing** processes performed as part of **resource** management—acquiring resources, developing the team, and managing the team

e. Discuss important **communications** concepts, and describe the process of managing communications

f. Describe the process of managing **stakeholder engagement**

g. Understand the process of **implementing risk** responses

h. Explain the process of **conducting procurements** and the need to effectively select sellers and create agreements

i. Summarize **leadership styles** and project management principles that assist in executing projects

j. Discuss unique aspects of executing agile/hybrid projects

EXECUTING PROJECT

THE IMPORTANCE OF EXECUTING PROJECT WELL

- Good plans are important, but it is even more **important to execute them well**.
  - In fact, *Fortune* magazine summarized research showing that the main reason chief executive officers (CEOs) failed was poor execution. Failed CEOs simply did not get things done, were indecisive, and did not deliver on commitments.
  - The same is true for project managers and all leaders.
- Stakeholders expect to see results from their projects through effective execution.
- In general, the **majority of a project's time and budget is spent** on project execution.
- Many of the deliverables and outputs created in the other process groups are fairly similar from project to project, but no **two projects are ever executed in the exact same way.**
  - Why? Because projects involve uncertainty or risk. No one can ever predict the challenges that project teams will face in trying to meet project goals.

SUMMARY OF EXECUTING PROCESSES AND OUTPUTS

Knowledge area	Executing process	Outputs
Project integration management	Direct and manage project work	Deliverables Work performance data Issue log Change requests Project management plan updates Project documents updates
	Manage project knowledge	Organizational process assets updates Lessons learned register Project management plan updates Organizational process assets updates
Project quality management	Manage quality	Quality report Test and evaluation documents Change requests Project management plan updates Project documents updates
Project resource management	Acquire resources	Physical resource assignments Project team assignments Resource calendars Change requests Project management plan updates Project documents updates Enterprise environmental factors updates
	Develop team	Organizational process assets updates Team performance assessments
	Manage team	Change requests Project management plan updates Project documents updates Enterprise environmental factors updates Organizational process assets updates Change requests Project management plan updates Project documents updates Enterprise environmental factors updates
Project communications management	Manage communications	Project communications Project management plan updates Project documents updates Organizational process assets updates
Project risk management	Implement risk responses	Change requests Project documents updates
Project procurement management	Conduct procurements	Selected sellers Agreements Change requests Project management plan updates Project documents updates Organizational process assets updates
Project stakeholder management	Manage stakeholder engagement	Change requests Project management plan updates Project documents updates Organizational process assets updates

PROJECT INTEGRATION MANAGEMENT

EXECUTING : PROJECT INTEGRATION MANAGEMENT

- During project execution, the project manager must direct and manage stakeholders to complete the project.
- ***Coordinate planning and execution:*** If the project manager and team did a good job planning, the plans will be easier to execute. As things change, team members need to update the plans to keep everyone working on the same page.
- ***Develop and use soft skills:*** for example, strong leadership, effective team building, strong communication, motivation, negotiation, conflict management, and problem solving. Project managers must lead by example in demonstrating the importance of creating good project plans and then following them in project execution.
- ***Provide a supportive organizational culture:*** Good project execution requires a supportive organizational culture.
- ***Break the rules when needed:*** the project manager must use his or her political skills to convince concerned stakeholders of the need to break the rules.
- ***Capitalize on product, business, and application area knowledge:*** Project managers should use their expertise to guide their team and make important decisions.
- ***Use project execution tools and techniques:*** The project management methodology should include guidelines on how to communicate project status, handle conflicts, work with suppliers and other stakeholders, and perform other important tasks.

DELIVERABLES

- Most project sponsors would say that the most important output of any project is its deliverables.
- Recall that deliverables are products or services produced or provided as part of a project.
  - They include product and process-related items.
  - Process-related deliverables related to developing and delivering those training materials and courses include researching existing training and meeting with potential partners.

WORK PERFORMANCE DATA

- One of a project manager's main jobs during project execution **was collecting, assessing, and communicating work performance data.**
- This data includes key performance indicators, technical performance measures, actuals (for start and end dates, durations, and costs), etc.
- In addition to reviewing this data, sometimes project managers use the "management by wandering around" (MBWA) approach, meaning they informally observe and talk to project team members and other stakeholders as much as possible.
- A common way to summarize work performance data is by using a **milestone** report.
  - Recall that a milestone is a significant point or event in a project, such as completing a major deliverable or awarding a major contract

ISSUE LOGS

- Many project managers keep an **issue log** to document, monitor, and track issues that need to be resolved for effective work to take place.
- An **issue** is a matter under question or dispute that could impede project success.
  - Issues could include situations in which people disagree, situations that need more clarification or investigation, or general concerns that need to be addressed.
- It is important to acknowledge issues that can hurt team performance and take action to resolve them.
- A critical issue is anything that prevents progress on scheduled activities.
  - If the activity is on the critical path, resolution is urgent.
  - There should be an issue escalation process that ensures issue resolution action.
  - For critical activities, resolution time limits and responsible roles must be defined.
  - For some issues, the steering committee may be the final resolution authority.

MILESTONE AND ISSUE LOGS EXAMPLE

Just-In-Time Training Project Milestone Report
September 1

Milestone	Date	Status	Responsible	Issues/Comments
Researched existing training	8/13	Complete	Jamie (replaced by Abner)	Many basic courses available, but not much advanced/tailored training. (Note: Replaced Jamie with better candidate for project after Jamie completed this task.)
Presented supplier management training survey results to steering committee	8/24	Complete	Kristin	Great feedback. Many people stressed the need to have instructor-led training and mentors for soft skills development.
Meetings with potential partners	9/21	In progress	Kristin/ Contracting	May need more time for meetings.

Issue Log as of October 1

Issue #	Issue Description	Impact on Project	Date Reported	Reported By	Assigned To	Priority (H/M/L)	Due Date	Status	Comments/ Follow-up
1	Key project team member is not working out	Can severely hurt project because Jamie is our supplier management expert	Aug 10	Kristin	Kristin	H	Oct 4	Open	Working with Jamie and appropriate managers to find a replacement
2	IT staff that is performing survey is over allocated	Delaying the survey will delay the entire project because it is a critical task	Sep 6	Mohamed	Kristin	H	Sep 28	Closed	Paid overtime was approved
Etc.									

CHANGE REQUESTS

- Often, several change requests emerge during project execution.
  - Recall that a process for handling changes should be defined during project planning as part of the project management plan.
  - Next chapter, monitoring and controlling projects, provides detailed information on handling changes.
- It is important during project execution to request appropriate changes formally and informally.
- Project managers, team members, suppliers, and other stakeholders can make change requests, so it is important to have a good process in place for handling them.
- Also recall that some projects require change management to address people's reactions when projects introduce big changes in organizations.

EXAMPLE PROJECT CHANGE REQUEST

Just-In-Time Training Project Change Request

Project Name: Just-In-Time Training Project
Date Request Submitted: September 22
Title of Change Request: Provide overtime to get survey results in time
Change Order Number: A200-17
Submitted by: Kristin Maur
Change Category:__Scope __Schedule X_ Cost __Technology __Other
Description of change requested:
In order to avoid a schedule slip and have appropriate internal resources available, we are requesting the approval of paid overtime for creating and distributing the survey for the supplier management course.
Events that made this change necessary or desirable:
The IT person has several other important projects on-hand. If these scheduled activities are delayed, the entire project will be delayed as they are on the critical path.
Justification for the change/why it is needed/desired to continue/complete the project:
We must send out and analyze the survey in a timely manner since we need the information to develop the first supplier management course and select an appropriate supplier.

Impact of the proposed change on:
Scope: None **Schedule:** None **Cost:** $550
Staffing: One IT person will work 10 hours of paid overtime basis over a period of several weeks.
Risk: Low. This person suggested the paid overtime and has successfully worked overtime in the past.
Other: None
Suggested implementation if the change request is approved:
Include the overtime pay in the normal paycheck.
Required approvals:

Name/Title	Date	Approve/Reject
Evan George/Affected Employee		
Stella Jacobs/Employee's Supervisor		
Julia Portman, VP of IT		

IMPLEMENTED SOLUTIONS TO PROBLEMS

Some common problems encountered during project execution include the following:

1. The project sponsor and/or other senior managers are not very supportive of the project.
2. Project stakeholders, such as people who would use the products and services the project is attempting to create, are not sufficiently involved in project decision-making.
3. The project manager is inexperienced in managing people, working in a particular organization, or understanding the application area of the project.
4. The project objectives/scope are unclear.
5. Estimates for time and cost goals are unreliable or unrealistic.
6. Business needs/technology changes have impacted the project.
7. People working on the project are incompetent or unmotivated
8. There are poor conflict-management procedures.
9. Communications are poor.
10. Suppliers are not delivering as promised.

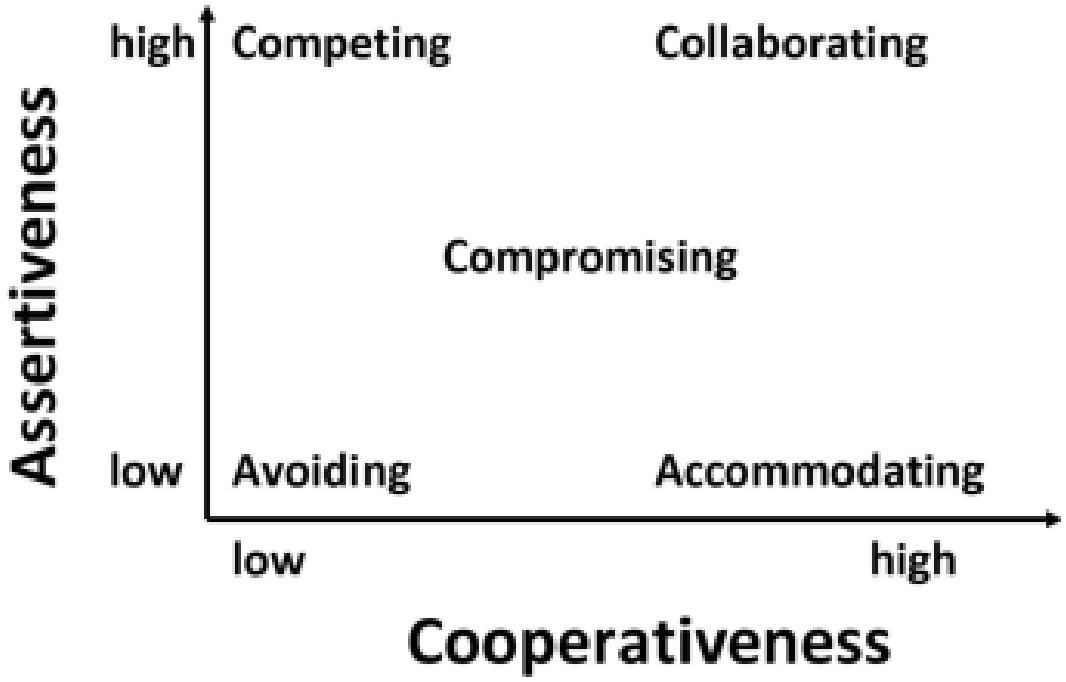
The first five problems should have been addressed during project initiation or planning (and were addressed in previous chapters), but they can also cause difficulties during execution.
Addressing business and technology changes are discussed in the next chapter on monitoring and controlling

CONFLICT MANAGEMENT

- Most large projects are high-stake endeavors that are highly visible within organizations. They require tremendous effort from team members, are expensive, require significant resources, and can have an extensive impact on the way work is done in an organization.
- When the stakes are high, conflict is never far away, and even small projects with low budgets have conflicts—it is a natural part of work and life in general.
- Project managers should lead their teams in developing norms for dealing with various types of conflicts that might arise.
- For example, team members should know that disrespectful behavior toward any project stakeholder is inappropriate, and that team members are expected to try to work out small conflicts themselves before elevating them to higher levels.
- The team charter, created during project planning, should address team conduct and conflict management.

CONFLICT HANDLING MODES

- The Thomas-Kilmann Instrument is designed to measure a person's behavior in conflict situations, defined as times when the concerns o people appear to be incompatible.
- In such conflict situations, you can describe an individual's behavior along two dimensions: (1) assertiveness, the extent to which people attempt to satisfy their own concerns, and (2) cooperativeness, the e to which people attempt to satisfy the other person's concerns.
- Project managers must also realize that not all conflict is bad.
- Conflict often produces important results, such as new ideas, better alternatives, and motivation to work harder and more collaboratively.
- Project team members might become stagnant or develop **groupthin** conformance to the values or ethical standards of a group—if there a conflicting viewpoints on various aspects of a project



MANAGING PROJECT KNOWLEDGE

- PMI added this new process to the PMBOK® Guide in 2017 to highlight the importance of managing project knowledge. There are two basic types of knowledge:
- **Explicit knowledge**: This type of knowledge can be easily explained using words, pictures, or numbers and is easy to communicate, store, and distribute. Examples include information found in textbooks and encyclopedias as well as project documents and plans.
- **Tacit knowledge**: Unlike explicit knowledge, tacit knowledge, sometimes called informal knowledge, is difficult to express and is highly personal. Examples include beliefs, insight, and experience. It is often shared through conversations and interactions between people. Many organizations set up programs like mentorships, communities of practice, or workshops to assist in passing on tacit knowledge.

- Knowledge management should be **done before, during, and after projects are completed**. It is often very difficult to accomplish. Organizations can provide explicit knowledge in writing or other formats, including audio, pictures, and videos. They can also build a culture of trust and sharing so people work together to pass on tacit knowledge. One of the main outputs of managing project knowledge is a lessons learned register.

LESSONS LEARNED REGISTER

- A lessons learned register should **document challenges, problems, realized risks and opportunities, and other** content to assist in knowledge management on current and **future** projects.
- It is best if project team members document lessons learned throughout the life of the project.
- At the end of the project, all lessons learned should be both discussed with the team and archived in a lessons learned repository, a key organizational process asset.

ID	Date identified	Owner	Name	Category	Situation	Recommendation
01	9/22	Kristin	Overtime	Cost	IT person was overbooked. Decided to pay overtime to finish survey work in time and not delay project end date.	Don't be afraid to suggest paid overtime, even though it's not used often at our organization.
02	10/1	Kim	Curriculum blog	Tech-nology	Kim was having trouble getting inputs on specific requirements for curriculum, so she set up a blog and provided incentives for inputs.	Encourage people to be creative. Allow use of some of the budget for incentives related to producing that deliverable.

Figure 7-6. Sample lessons learned register

MANAGING QUALITY

MANAGING QUALITY

- It is one thing to develop a plan for ensuring quality on a project; it is another to ensure delivery of quality products and services.
- The term **quality assurance** is often used to describe the activities related to satisfying the relevant quality standards for a project.
  - It addresses the following question: Are we doing what we said we would do, when we said we would do it, in the way we said we would do it, and can we do it better?
- Another goal of managing quality is continual quality improvement.
- Key outputs of managing quality include a quality report, test and evaluation documents, change requests, project management plan updates, and project documents updates.

QUALITY IMPROVEMENT TOOLS AND TECHNIQUES

- **Benchmarking** generates ideas for quality improvements by comparing specific project practices or product characteristics to those of other projects or products within or outside of the organization itself.
- A **quality audit** is a structured review of specific quality management activities that helps identify lessons learned, which could improve performance on current or future projects.
  - In-house auditors or third parties with expertise in specific areas can perform quality audits, which can be either scheduled or random.
  - A quality audit could be done periodically to review the survey results.
- **Process analysis** involves analyzing how a process operates and determining improvements. There are several ways to perform process analysis. You can use a simple task board or **kanban board** to visually display work in columns labeled To Do, In Progress, and Done. Root cause analysis is also often part of process analysis.
  - **Cause-and-effect diagrams**—also called fishbone diagrams (because their structure resembles a fishbone) or Ishikawa diagrams (named after their creator)—can assist in ensuring and improving quality by finding the root causes of quality problems.

SAMPLE KANBAN BOARD & CAUSE-EFFECT DIAGRAM

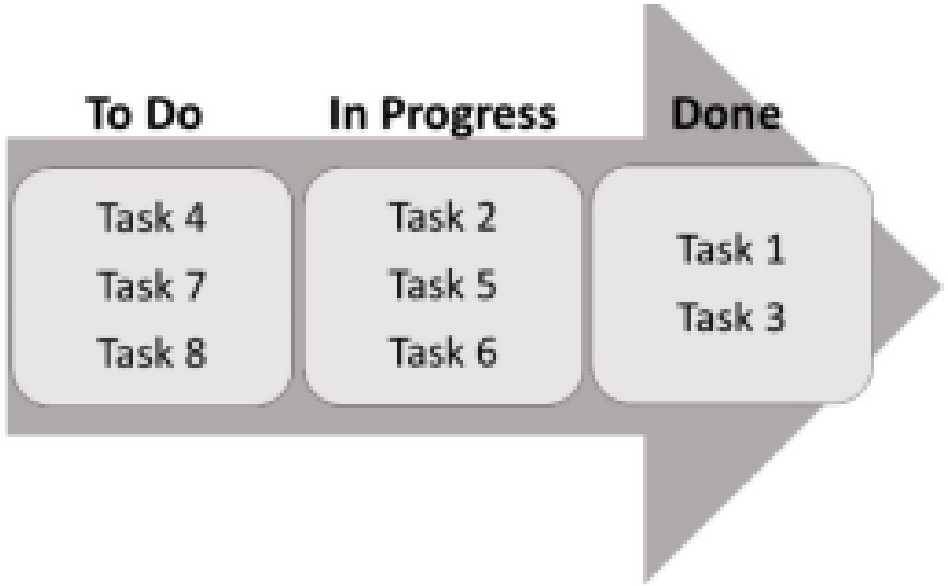
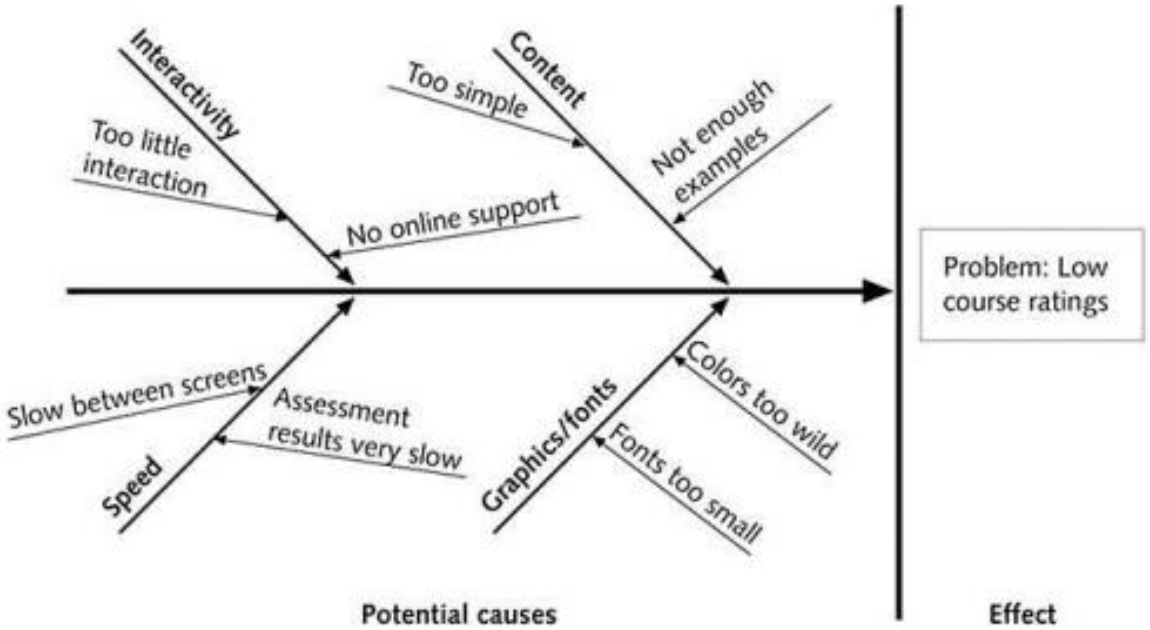


Figure 7-7. Sample kanban board



PROJECT RESOURCE MANAGEMENT

PROJECT RESOURCE MANAGEMENT

- Effective use of resources is crucial to project execution.
- The main processes project managers perform include acquiring resources, developing the team, and managing the team.
- Key outputs include physical resource assignments, project team assignments, resource calendars, team performance assessment, change requests, and updates to the project management plan, enterprise environmental factors, and organizational process assets.
- Resource calendars are simply calendars for each resource showing work assignment dates.
- Before discussing resource assignments and team performance assessment, it is important to understand basic concepts related to dealing with people in a work setting. Key concepts include motivation, influence, and effectiveness.

MOTIVATION

- Psychologists, managers, coworkers, teachers, parents, and most people in general still struggle to understand what motivates people, or why they do what they do.
- **Intrinsic motivation** causes people to participate in an activity for their own enjoyment. For example, some people love to read, write, or play an instrument because it makes them feel good.
- **Extrinsic motivation** causes people to do something for a reward or to avoid a penalty. For example, some young children would prefer *not* to play an instrument, but do so to receive a reward or avoid a punishment.

MASLOW'S HIERARCHY OF NEEDS

- The basic pyramid structure of **Maslow's hierarchy of needs**, which states that people's behaviors are guided or motivated by a *sequence* of needs.
- At the bottom of the hierarchy are physiological needs, such as air, water, and food.
- After physiological needs are satisfied, safety needs— such as shelter from bad weather, lack of physical or mental abuse, and a lowcrime environment—guide behavior.
- After safety needs are satisfied, social needs—such as having friends, belonging to groups, and having a sense of community—come to the forefront, and so on up the hierarchy.
- Examples of esteem needs include personal achievement, recognition, and respect, whereas self-actualization needs include a sense of fulfillment and belief that one is working to his or her potential.
- The order of these needs in the pyramid is significant. Maslow suggests that each level of the hierarchy is a prerequisite for the level above

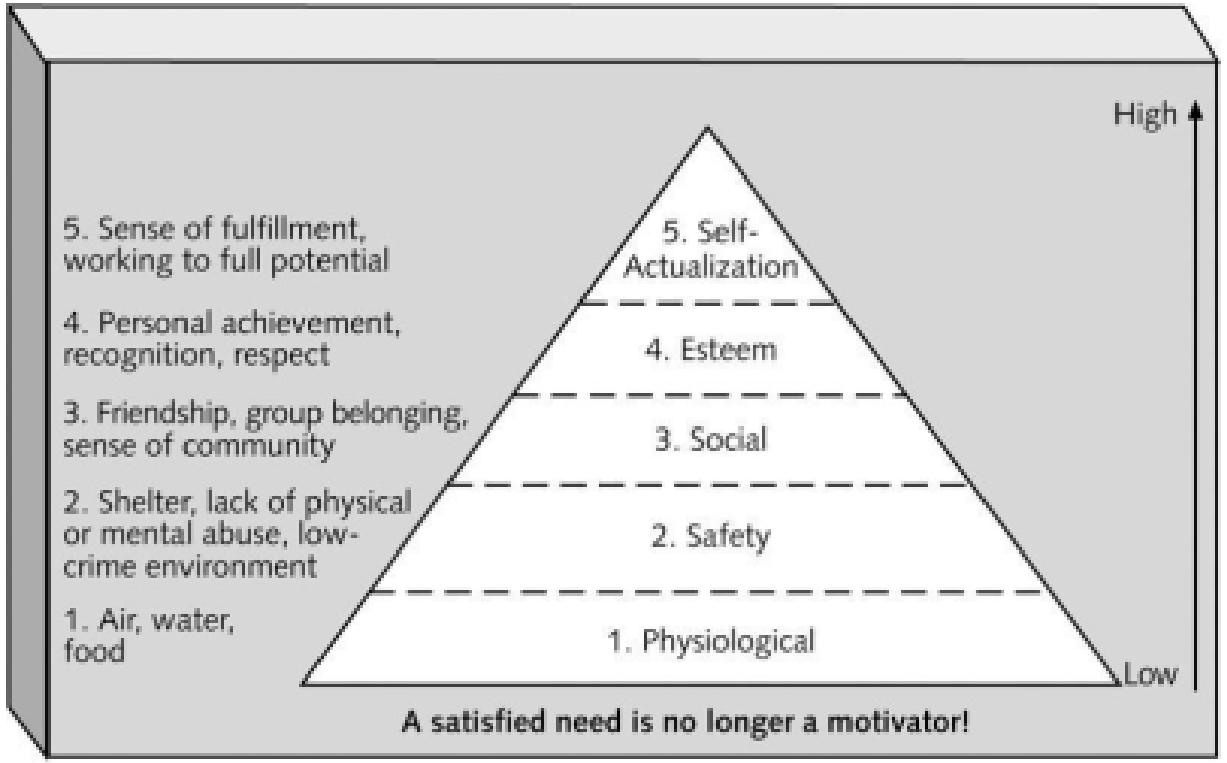


Figure 7-10. Maslow's hierarchy of needs (Schwalbe, Information Technology Project Management, Sixth Edition, 2010)

HERZBERG'S MOTIVATION-HYGIENE THEORY

- Frederick Herzberg, a psychologist and professor, is best known for distinguishing between motivational factors and hygiene factors when considering motivation in work settings.
- He called factors that cause job satisfaction **motivators**, and factors that cause dissatisfaction **hygiene** factors.
  - A hygiene factor is a basic necessity, such as proper heating or cooling during extreme weather.
  - Hygiene factors would cause dissatisfaction if not present but would not motivate workers to do more if present.
- Herzberg found that people were **motivated** to work mainly by feelings of personal achievement and recognition.
  - **Motivators**, Herzberg concluded, included achievement, recognition, the work itself, responsibility, advancement, and growth.

MCCLELLAND'S ACQUIRED-NEEDS THEORY

The main categories of acquired needs include achievement, affiliation, and power. Normally, one or two of these needs is dominant in individuals.

- *Achievement*: People with a high need for achievement (nAch) seek to excel and tend to avoid both low-risk and high-risk situations to improve their chances of achieving something worthwhile.
  - Achievers need regular feedback and often prefer to work alone or with other high achievers. Managers should give high achievers challenging projects with achievable goals. Achievers should receive frequent performance feedback, and although money is not an important motivator to them, it is an effective form of feedback as a way to keep score.
- *Affiliation*: People with a high need for affiliation (nAff) desire harmonious relationships with other people and need to feel accepted by others.
  - They tend to conform to the norms of their work group and prefer work that involves significant personal interaction. Managers should try to create a cooperative work environment to meet the desires of people with a high need for affiliation.
- *Power*: People with a need for power (nPow) desire either personal power or institutional power.
  - People who need personal power want to direct others and can be seen as bossy. People who need institutional, or social, power want to organize others to further the goals of the organization. Management should provide those seeking institutional power with the opportunity to manage others, emphasizing the importance of meeting organizational goals.

MCGREGOR'S THEORY X AND THEORY Y AND OUCHI'S THEORY Z

McGregor found that although many managers spouted the right ideas, they actually followed a set of assumptions about worker motivation that he called Theory X (sometimes referred to as classical systems theory).

- **People who believe in Theory X** assume that workers dislike and avoid work if possible, so managers must use coercion, threats, and various control schemes to get workers to make adequate efforts to meet objectives. Theory X managers assume that the average worker wants to be directed and prefers to avoid responsibility, has little ambition, and wants security above all else.
- **Managers who believe in Theory Y** assume that individuals do not inherently dislike work but consider it as natural as play or rest. The most significant rewards are the satisfaction of esteem and self-actualization needs, as described by Maslow. McGregor urged managers to motivate people based on these more valid Theory Y notions.

William Ouchi, author of *Theory Z* (1981), developed this theory after making a comparative study of Japanese and American management practices.

- **Theory Z** emphasizes motivating workers and increasing loyalty by providing **a job for life** with a strong focus on the well-being of the employee, both on and off the job. Although the concept of a job for life is not very popular today, many people reassessed their views of work-life balance during the COVID pandemic, as described in the following Media Snapshot.

THAMHAIN AND WILEMON'S INFLUENCE BASES

Many people working on a project do not report directly to project managers, and project managers often do not have control over project staff that report to them.

1. ***Authority:*** The legitimate hierarchical right to issue orders
2. ***Assignment:*** The project manager's perceived ability to influence a worker's assignment to future projects
3. ***Budget:*** The project manager's perceived ability to authorize the use of discretionary funds
4. ***Promotion:*** The ability to improve a worker's position
5. ***Money:*** The ability to increase a worker's pay and benefits
6. ***Penalty:*** The project manager's perceived ability to dispense or cause punishment
7. ***Work challenge:*** The ability to assign work that capitalizes on a worker's enjoyment of doing a particular task, which taps an intrinsic motivational factor
8. ***Expertise:*** The project manager's perceived specialized knowledge that others deem important
9. ***Friendship:*** The ability to establish friendly personal relationships between the project manager and others

Projects were **more likely to fail** when project managers relied too heavily on using *authority, money, or penalty* to influence people.

When project managers used *work challenge and expertise* to influence people, projects were more **likely to succeed**

COVEY'S EFFECTIVENESS

Covey's first three habits of effective people—first three habits—help people achieve a private victory by becoming independent. After achieving independence, people can then strive for interdependence by developing the next three habits and synergize. **Synergy** is the concept that the whole is equal to more than the sum of its parts. Finally, seventh habit to develop and renew their physical, spiritual, mental, and social/emotional selves.

1. *Be proactive:* Project managers must be proactive, anticipate, and plan for problems and inevitable changes on projects.
2. *Begin with the end in mind:* The project manager must have and share the vision of the final outcome of the project.
3. *Put first things first:* Project managers should focus on important and not urgent activities, such as developing various project plans, building relationships with major project stakeholders, and mentoring project team members.
4. *Think win/win:* Project managers should strive to use a win/win approach in making decisions, but sometimes, especially in competitive situations, they must use a win/lose paradigm
5. *Seek first to understand, then to be understood:* **Empathic listening** is listening with the intent to understand by putting yourself in the shoes of the other person.
6. *Synergize:* In projects, a project team can synergize by creating collaborative products that are much better than a collection of individual efforts
7. *Sharpen the saw:* Project managers must make sure that they themselves and their project team have time to retrain, reenergize, and occasionally even relax to avoid burnout.

ACQUIRING RESOURCES

- Project resources include both physical and human resources.
  - Physical resources include equipment, supplies, and materials.
  - Human resources include the people required to perform the project work.
- There's a saying that the project manager who is the smartest person on the team has done a poor job of recruiting!
- Project managers with strong influencing and negotiating skills are often good at getting internal people to work on their project.
- However, the organization must ensure that people assigned to the project **best fit the project's** requirements, and that these people are motivated to remain on the project.
- Problems that often occur when assigning resources are **availability** and **overallocation**. You can use ***resource loading and leveling*** to help address these problems.

RESOURCE LOADING

- **Resource loading** refers to the number of individual resources an existing schedule requires during specific time periods.
- Resource loading helps project managers develop a general understanding of the demands a project will make on the organization's resources (physical and human), as well as on individual people's schedules.
- A histogram can be very helpful in determining staffing needs or in identifying staffing problems.
- A resource histogram can also show when work is being overallocated to a certain resource or group.
- **Overallocation** means more resources than are available are assigned to perform work at a given time

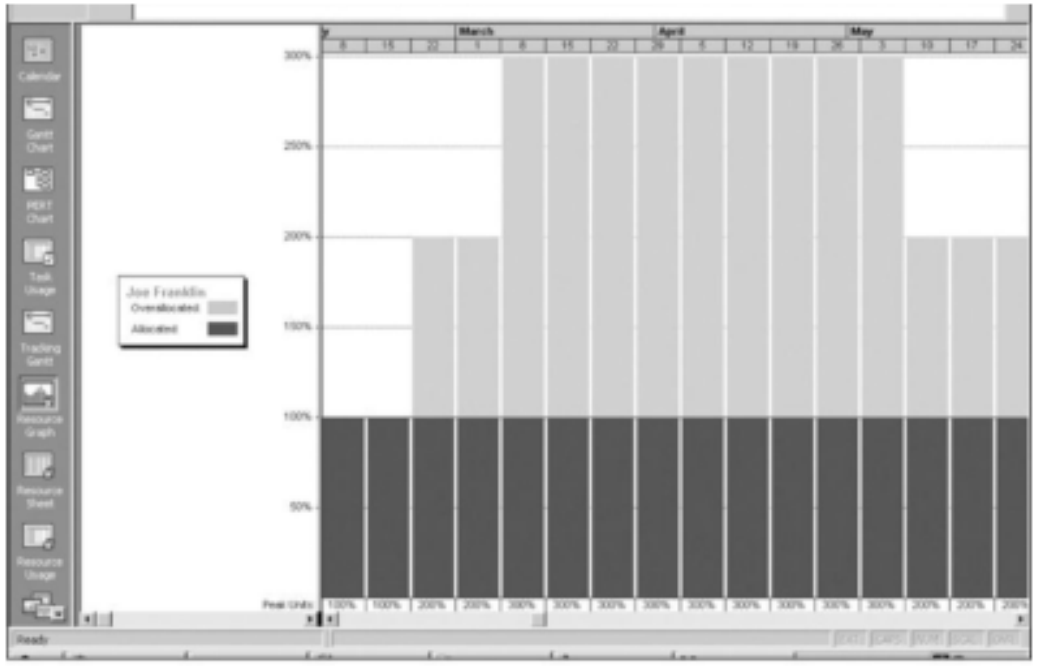
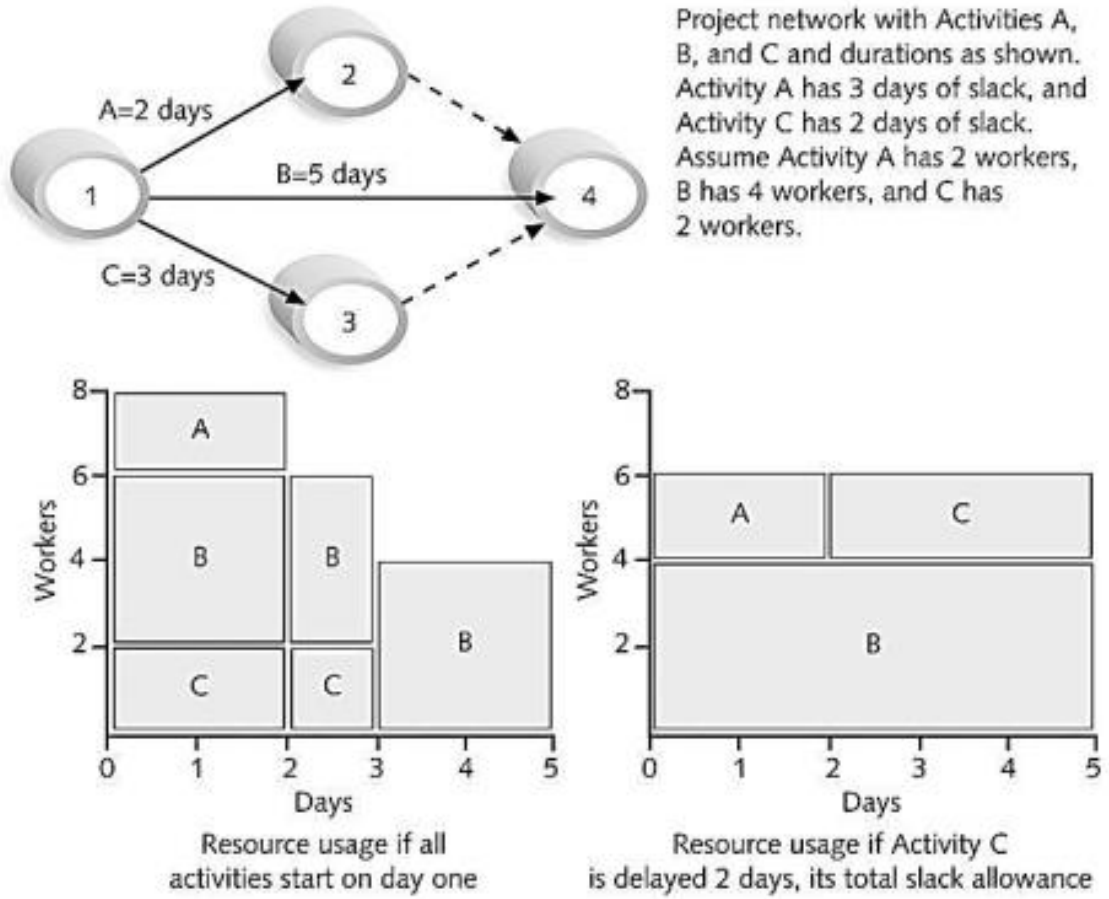


Figure 7-11. Sample resource histogram showing an overallocated individual

RESOURCE LEVELING

- **Resource leveling** is a technique for resolving resource conflicts by delaying tasks.
- It is a form of network analysis in which resource management concerns drive scheduling decisions (start and finish dates).
- The main purpose of resource leveling is to create a smoother distribution of resource usage.
- Project managers examine the network diagram for areas of slack or float, and to identify resource conflicts.
- Resource leveling, therefore, aims to minimize period-by-period variations in resource loading by shifting tasks within their slack or float allowances.



RESOURCE LEVELING ADVANTAGE

- When resources are used on a more constant basis, they **require less management**.
- Resource leveling may enable project managers to use **a just-in-time inventory** type of policy for using subcontractors or other expensive resources.
- Resource leveling results in **fewer problems for project personnel and accounting departments**. Increasing and decreasing labor levels and human resources often produce additional work and confusion.
- Resource leveling often **improves morale**. People like to have some stability in their jobs.

Resource smoothing, also known as time constrained scheduling, is a technique you can use to finish a project on time while avoiding peaks and valleys in resource demand.

SAMPLE RESOURCE ASSIGNMENTS

Sample Physical Resource Assignments

Physical Resource	Date(s) Reserved	Description
Classroom 1	1/12	Need this room for executive course
Classroom 2	12/15, 1/20, 1/26, 2/3, 2/9	Need this room or Classroom 4
Classroom 3	2/3	Need this room for software applications course – has required student computers

Figure 7-13. Sample physical resource assignments

Sample Project Team Assignments

Team Roster
September 1

Project Name: Just-In-Time Training Project

Name	Role on Project	Position	Email	Phone	Location
Mike Sundby	Project Champion	VP of HR	msundby@ globalconstruction.com		
Lucy Camerena	Project Sponsor	Training Director	lcamerena@ globalconstruction.com		
Kristin Maur	Project Manager	Project Manager	kmaur@ globalconstruction.com		
Mohamed Abdul	Team Member	Senior programmer/ analyst	mabdul@ globalconstruction.com		
Kim Johnson	Team Member	Curriculum designer	kjohnson@ global construction.com		
Abner Tomas	Team Member	Supplier management expert	atomas@ globalconstruction.com		

Figure 7-14. Sample team roster

DEVELOPING THE PROJECT TEAM

- The main goal of **team development** is to help people work together more effectively to improve project performance
- It takes teamwork to successfully complete most projects

TUCKMAN MODEL OF TEAM DEVELOPMENT

1. **Forming**: introduction of team members
2. **Storming**: team members have different opinions as to how the team should operate
3. **Norming**: develop a common working method
4. **Performing**: emphasis on reaching the teams goal rather than working on team process
5. **Adjourning**: break-up the team after they successfully reach their goals.

TRAINING

- Training can help people understand themselves, each other, and how to work better in teams
- Team building activities include:
  - Physical challenges
  - Psychological preference indicator tools

MEYERS-BRIGGS TYPE INDICATOR (MBTI)

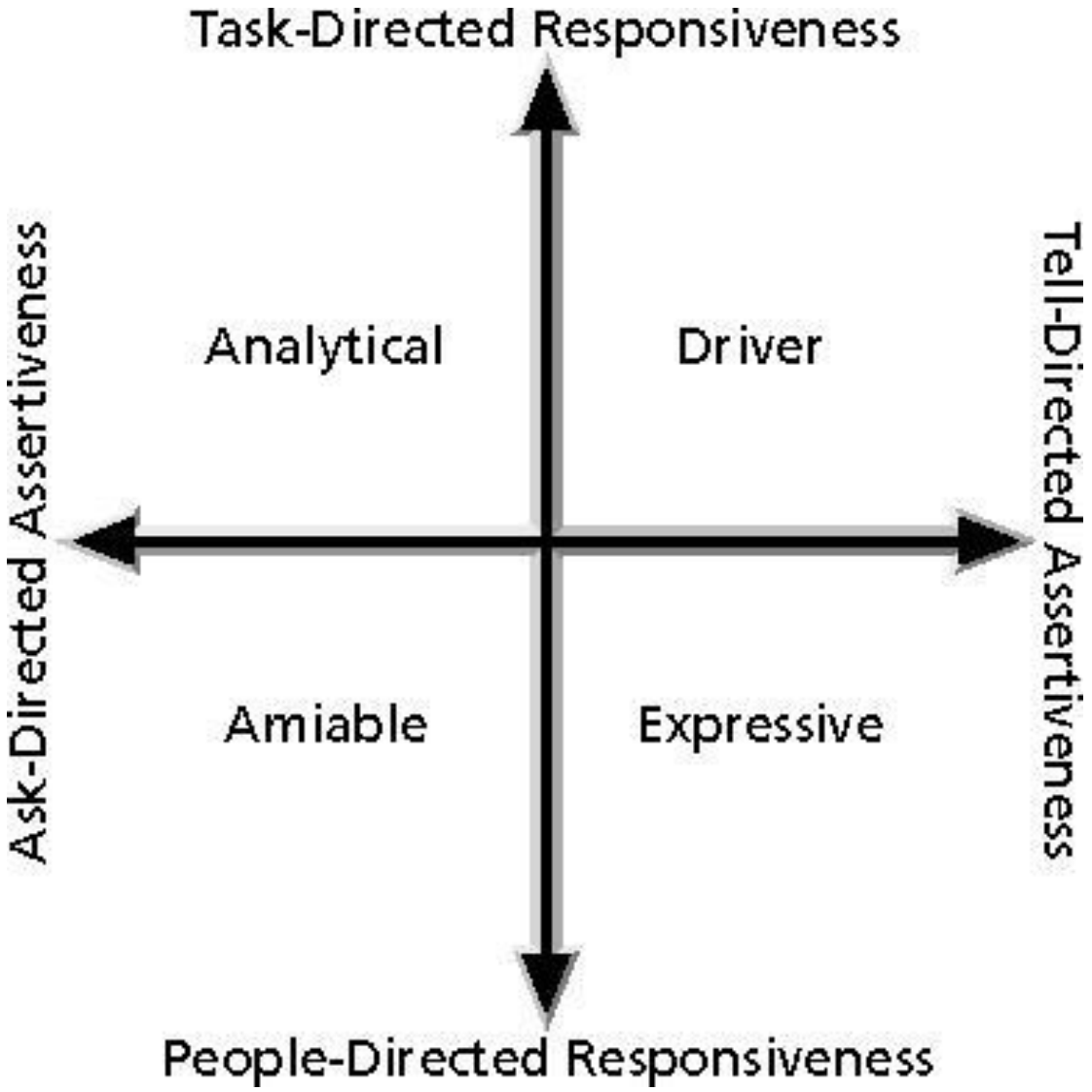
- MBTI is a popular tool for determining personality preferences and helping teammates understand each other
- Four dimensions include:
  - Extrovert/Introvert (E/I)
  - Sensation/Intuition (S/N): gather information, facts vs intuitive
  - Thinking/Feeling (T/F): judgment, objective & logical vs subjective & personal
  - Judgment/Perception (J/P): attitude, structure vs flexible

IT people vary most from the general population in not being extroverted or sensing

- **75% IT people Introvert, general population 25% introvert**
- **55% IT people Intuitive, general population 25%**
- **80% IT people Thinking, general population 50%**

SOCIAL STYLES PROFILE

- People are perceived as behaving primarily in one of four zones, based on their assertiveness and responsiveness:
  - **Drivers**: pro active & task oriented
  - **Expressives**: proactive and people oriented
  - **Analyticals**: reactive & task oriented
  - **Amiables**: reactive & people oriented
- People on opposite corners (drivers and amiables, analyticals and expressives) may have difficulties getting along



REWARD AND RECOGNITION SYSTEMS

- **Team-based reward** and recognition systems can promote teamwork
- Focus on **rewarding teams** for achieving specific goals
- Allow time **for team members to mentor and help each other** to meet project goals and develop human resources

SAMPLE TEAM PERFORMANCE ASSESSMENT

- Project managers assess team performance in several different ways.
- Note that the term performance assessment is used for project teams while performance appraisal is for individuals.

Team Performance Assessment
August 1

Project Name: Just-In-Time Training Project

Individual's Name: _____ **Project Manager:** Kristin Maur

Date: _____

1. Using a scale of 0-100, assess how you think **the project team** is performing: ___
2. Explain the rationale behind the above score.
3. Using a scale of 0-100, assess how **you** are performing on this project: _____
4. Explain the rationale behind the above score. What are your roles and responsibilities, and how well have you performed them?
5. Briefly assess each team member's performance. If you had to give each person a score between 0-100, what would it be?
6. To compare individual contributions, if you had 100 points to allocate to your team, how would you allocate them?
7. What suggestions do you have for improving team performance?

Figure 7-16. Sample team performance assessment

MANAGING THE PROJECT TEAM

- Project managers must **lead** their teams in performing various project activities
- After assessing team **performance** and related information, the project manager must decide:
  - If changes should be requested to the project
  - If **corrective** or **preventive** actions should be recommended
  - If updates are needed to the project management plan or organizational process assets

APPROACHES FOR MANAGING PROJECT TEAMS

- Observation and conversation
- Feedback
- Conflict management

GENERAL ADVICE ON TEAMS

- Be patient and kind with your team
- Fix the problem instead of blaming people
- Establish regular, effective meetings
- Allow time for teams to go through the basic team-building stages
- Limit the size of work teams to three to seven members
- Plan some social activities to help project team members and other stakeholders get to know each other better
- Stress team identity
- Nurture team members and encourage them to help each other
- Take additional actions to work with virtual team members

ADDITIONAL ADVICE FOR MANAGING VIRTUAL TEAMS

1. Hire the right people. If your team is not in place yet, be sure to hire people with strong professional maturity who can work on projects autonomously. If the team is already in place, take the time to address needs and concerns about working virtually.
2. Trust your people. Do not micromanage or try to monitor how they are spending their time. Focus on people completing work, not how or when they complete it.
3. Tailor communications. Understand how each team member prefers to communicate. Also let them know how you like to communicate.
4. Use technology effectively. Decide what tools to use when, and provide training as needed. Establish rules for videoconferencing, like having your video on.
5. Use rituals. Planning the week on Monday mornings, reviewing accomplishments and previewing the week ahead on Fridays, and similar rituals help teams and their leaders.
6. Own your own schedule, and let people own theirs. Make sure meetings are meaningful. Use other tools like email for general topics and project management software to enter and track progress.
7. Make the time to get personal. Build connections by finding out what team members like to do outside of work.
8. Build a strong team culture. Work to build a sense of team using virtual meetings or events like virtual runs/walks.
9. Be available. Check in with team members to see how they are doing.
10. Be unavailable. Take time for yourself! Take time off when you need to, and let team members know they should do the same.

PROJECT COMMUNICATIONS MANAGEMENT

IMPORTANT PROJECT COMMUNICATIONS CONCEPTS

Key concepts include:

- formal and informal communications,
- nonverbal communications,
- using the appropriate communications medium,
- understanding individual and group communication needs, and
- the impact of team size on project communications.

IMPORTANCE OF FACE-TO-FACE COMMUNICATION

- Research says that in a face-to-face interaction:
  - 58 percent of communication is through body language.
  - 35 percent of communication is through how the words are said
  - 7 percent of communication is through the content or words that are spoken
- Pay attention to more than just the actual words someone is saying
- A person's tone of voice and body language say a lot about how he or she really feels

PERSONAL PREFERENCES AFFECT COMMUNICATION NEEDS

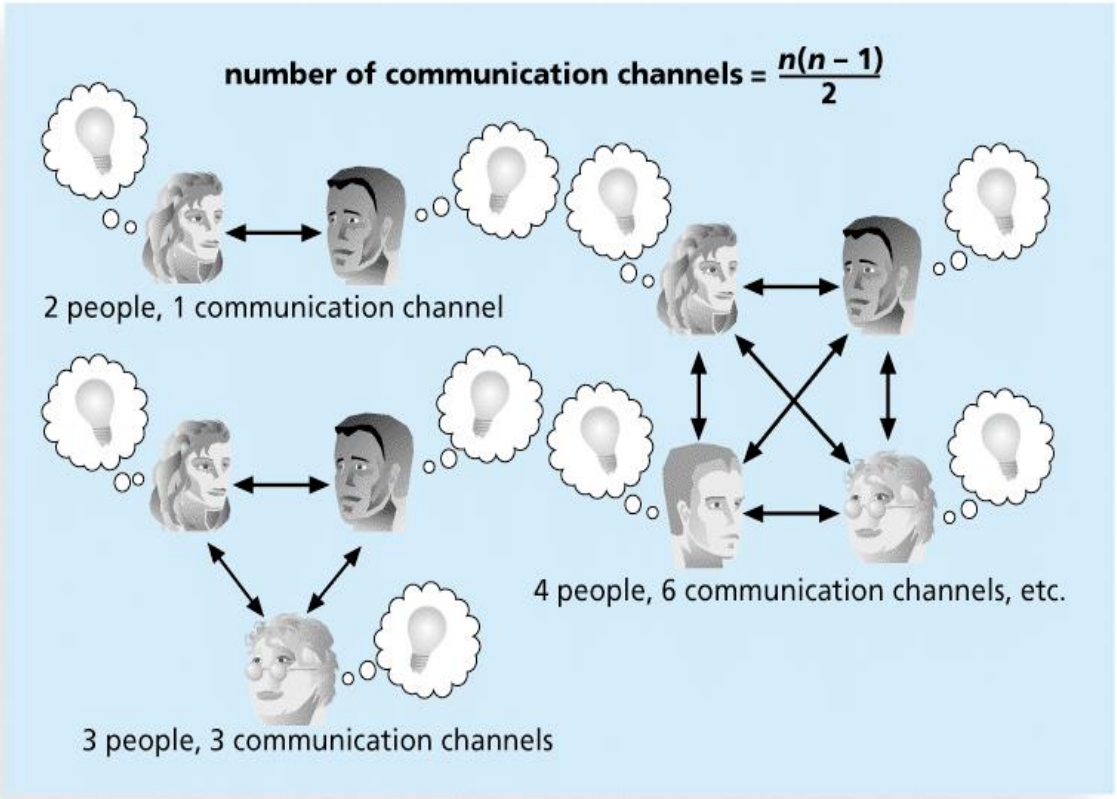
- **Introverts** like more private communications, while **extroverts** like to discuss things in public
- **Intuitive** people like to understand the big picture, while **sensing** people need step-by-step details
- **Thinkers** want to know the logic behind decisions, while **feeling** people want to know how something affects them personally
- **Judging** people are driven to meet deadlines while **perceiving** people need more help in developing and following plans

- Encouraging More Face-to-Face Interaction
  - Short, frequent meetings are often very effective in IT projects
  - Stand-up meetings force people to focus on what they really need to communication
  - Some companies have policies preventing the use of e-mail between certain hours or even entire days of the week

DISTRIBUTING INFORMATION IN AN EFFECTIVE AND TIMELY MANNER

- Don’t bury crucial information
- Don’t be afraid to report bad information
- Oral communication via meetings and informal talks helps bring important information—good and bad—out into the open
- Other Communication Considerations
  - Rarely does the receiver interpret a message exactly as the sender intended
  - Geographic location and cultural background affect the complexity of project communications
    - Different working hours
    - Language barriers
    - Different cultural norms

DETERMINING THE NUMBER OF COMMUNICATIONS CHANNELS



- As the number of people involved increases, the complexity of communications increases because there are more communications channels or pathways through which people can communicate.
- Number of communications channels = $\frac{n(n-1)}{2}$

  where *n* is the number of people involved

PROJECT COMMUNICATIONS AND UPDATING BUSINESS PROCESSES

- Getting project information to the right people at the right time and in a useful format is just as important as developing the information in the first place.
- The communications management plan prepared during project planning serves as a good starting point for project communications.
- During execution, project teams must address important considerations for creating and distributing project communication, as described previously.
- One of the outputs of managing communications is organizational process assets updates, such as improving business processes.

PROJECT STAKEHOLDER MANAGEMENT

MANAGING STAKEHOLDER ENGAGEMENT

- It is important for project managers to engage their stakeholders to help their projects succeed.
  - The stakeholder management plan should identify key stakeholders and describe strategies for managing relationships with them.
  - It is easy to say that all project managers have to do is follow their plans.
- Managing stakeholder engagement, like managing student engagement, can be extremely difficult.
  - It is important to set the stage early that stakeholder engagement is expected and welcomed.
- A project manager's leadership and interpersonal skills are crucial in making that happen.
- Inputs to managing stakeholder engagement include the project management plan, (especially the sections regarding stakeholder engagement, communications, and risk), project documents (i.e., change log, issue log, lessons learned register, and stakeholder register), enterprise environmental factors, and organizational process assets.
  - Many of the outputs are similar to other knowledge areas, such as change requests and updates to the project management plan and project documents.
- Tools and techniques include expert judgment, inspection, communication skills, interpersonal and team skills, ground rules, and meetings.

PROJECT RISK MANAGEMENT

IMPLEMENTING RISK RESPONSES

- The main executing process performed as part of project risk management is implementing risk responses as defined in the process to plan risk responses.
- Key outputs include change requests and project documents updates (i.e., issue log, lessons learned register, project team assignments, risk register, and risk report)
- It is important to plan risk responses, and it is even more important to implement them well.
-

PROJECT PROCUREMENT MANAGEMENT

CONDUCTING PROCUREMENTS

- ***Approaching a preferred supplier:*** Many organizations have formed good working relationships with certain suppliers, so they want to continue working with them.
- ***Approaching several qualified suppliers:*** If it does not get the desired response, the buyer can either expand its list of potential sellers until it gets the desired response or revise its procurement plans.
- ***Advertising to many potential suppliers:*** As a result of pursuing a competitive bidding strategy, the buyer can receive better goods and services than expected at a lower price or with faster delivery. **Advertising the procurement** (on a website, in a trade journal, or by other means) and receiving proposals and bids from multiple sources often takes advantage of the competitive business environment.

SAMPLE QUALIFIED SELLER LIST

- Selecting suppliers or sellers, often called source selection, involves:
  - evaluating proposals or bids from sellers,
  - choosing the best one,
  - negotiating the contract, and
  - awarding the contract.
- Several stakeholders in the procurement process should be involved in selecting the best suppliers for the project.
- Often, teams of people are responsible for evaluating various sections of the proposals.
- Often, buyers develop a **short list** of the top three to five suppliers to reduce the work involved in selecting a source.
- It is customary to conduct contract negotiations during the source selection process.
- Sellers on the short list are often asked to prepare a best and final offer (BAFO).
  - Expert negotiators often conduct these negotiations, especially for contracts involving large amounts of money. In addition, senior managers from both buying and selling organizations often meet before making final decisions.

Qualified Sellers List – September 9

Project Name: Just-In-Time Training Project

Seller Name/ Website	Areas of Expertise	Full-Time Staff	Reputation
Company A www.coA.com	Construction industry, supplier management, project management	40	One of few training firms that specializes in training for the construction industry
Company B www.coB.com	E-learning, custom course development	100	Has many partnerships with other companies, reasonable prices
Company C www.coC.com	Project management, negotiating skills	10	Small firm but well respected, does instructor-led and e-learning
Etc.			

SAMPLE AGREEMENT OR CONTRACT

Global Construction, Inc.
Service Agreement – August 10

Title of Work: Qualified Sellers List and Report

This is an Agreement made as of _____ by ABC Training Consultants, 2255 River Road, Boston, MA (the "Seller"), and Global Construction, Inc., 5000 Industrial Drive, Minneapolis, MN (the "Buyer").

THE SELLER AND THE BUYER AGREE THAT:

1. The Work: The Seller will create the Work as set forth in Exhibit A hereto. The Buyer will provide the Seller with the format and specifications in which each element of the Work is to be submitted. The Seller agrees to conform to such format and specifications.
2. Delivery of the Work: The Seller agrees to deliver to the Buyer the Work in form and content acceptable to the Buyer on or before the dates outlined in Exhibit B of this Agreement, time being of the essence to the Buyer.
3. Right to Terminate: If the Seller materially departs from the agreed-upon schedule or if the Work is not satisfactory to the Buyer (based on reviews of drafts, market conditions, and/or other criteria as determined by the Buyer), the Buyer may at its option:
   A. Allow the Seller to finish, correct, or improve the Work by a date specified by the Buyer;
   B. Terminate this Agreement by giving written notice to the Seller.
4. Payments: The Buyer will pay the Seller a fixed price of $5,000 upon accepted completion of the Work.
5. Exhibit: The following Exhibit is hereby incorporated by reference into this Agreement:

Exhibit A: Statement of Work
Exhibit B: Schedule

IN WITNESS WHEREOF, THE PARTIES HERETO HAVE EXECUTED THIS Agreement as a sealed instrument as of the date first above written.

	Global Construction, Inc.	ABC Training Consultants
By:	_____	_____
Date	_____	_____

Figure 7-18. Sample agreement or contract

APPLYING LEADERSHIP STYLES AND PROJECT MANAGEMENT PRINCIPLES TO PROJECT EXECUTION

LEADERSHIP STYLES

The *PMBOK® Guide – Sixth Edition* briefly describes the following leadership styles:

1. **Laissez-faire**: Meaning "let go," this hands-off approach lets teams determine their own goals and how to achieve them.
2. **Transactional**: This management by exception approach focuses on achieving goals or compliance by offering team members appropriate rewards and punishments.
3. **Servant leader**: People using this approach focus on other people's growth, learning, and autonomy and demonstrate service as a form of leadership.
4. **Transformational**: By working with others to identify needed changes, these leaders empower others and guide changes through inspiration.
5. **Charismatic:** These people can inspire others based on their enthusiasm and confidence.
6. **Interactional**: This leadership style is a combination of transactional, transformational, and charismatic.

OTHER LEADERSHIP STYLE

Daniel Goleman, author of *Emotional Intelligence* (1995), also wrote a book called *Primal Leadership: Unleashing the Power of Emotional Intelligence* (2013), which describes six different styles of leadership and situations where they are most appropriate:

1. ***Visionary***: Needed when an organization needs a new direction, and the goal is to move people towards a new set of shared dreams. The leader articulates where the group is going, but lets them decide how to get there by being free to innovate, experiment, and take calculated risks.
2. ***Coaching***: One-on-one style that focuses on developing individuals, showing them how to improve their performance. This approach works best with workers who show initiative and request assistance.
3. ***Affiliative***: Emphasizes the importance of teamwork and creating harmony by connecting people to each other. This approach is effective when trying to increase morale, improve communication, or repair broken trust.
4. ***Democratic***: Focuses on people's knowledge and skills and creates a commitment to reaching shared goals. This leadership style works best when the leader needs the collective wisdom of the group to decide on the best direction to take for the organization.
5. ***Pacesetting***: Used to set high standards for performance. The leader wants work to be done better and faster and expects everyone to put forth their best effort.
6. ***Commanding***: Most often used, also called autocratic or military style leadership. This style is most effective in a crisis or when a turnaround is needed.

APPLYING PROJECT MANAGEMENT PRINCIPLES RELATED TO EXECUTING PROJECTS

Several project management principles can assist in executing projects:

1. **Demonstrate leadership behaviors.**
2. **Be a diligent, respectful, and caring steward.** "Stewards act responsibly to carry out activities with integrity, care, and trustworthiness while maintaining compliance with internal and external guidelines.
3. **Effectively engage with stakeholders.** By managing stakeholder engagement, people are more likely to be engaged in projects. Projects are done by and for people,so it's crucial to effectively engage with stakeholders.
4. **Recognize, evaluate, and respond to system interactions.** Projects are not run in isolation. Project managers and their teams need to take a holistic view of their projects, remembering that they are part of a larger system, especially during execution.

EXECUTING AGILE/HYBRID PROJECTS

EXECUTING AGILE/HYBRID PROJECTS

- Agile/hybrid project managers and their teams **can use any of the predictive project executing processes**, tools, or techniques mentioned earlier.
  - For example, they still direct and manage project work, manage project knowledge, manage quality, acquire resources, develop teams, manage teams, manage communications, implement risk responses, conduct procurements, and manage stakeholder engagement.
- An important characteristic of agile teams is that they **should be self-directed**.
- The Scrum master or project manager, if there is one, should primarily use a **servant-leader approach**, focusing on other people's growth, learning, and autonomy.
- Agile teams **can face the same problems addressed earlier** (unsupportive project sponsors, unengaged stakeholders, poor communications, etc.).

THEORY VERSUS PRACTICE

- Theory:
  - that there are dozens of ways to run projects using an agile approach, with Scrum being the most popular.
  - Also Scrum does not include a project manager role.
  - Scrum teams have one product owner, one Scrum master, and several developers.
  - According to The Scrum Guide, “Scrum Teams are cross-functional, meaning the members have all the skills necessary to create value each Sprint. They are also self-managing, meaning they internally decide who does what, when, and how.”
  - The Scrum master, theoretically, would not assign tasks to individual people because the team is supposed to work together to decide who does what.
  - The Scrum master would also remove any roadblocks the team faces.
  - The product owner would share the vision of the project and provide the prioritized product backlog to ensure the team produces valuable work.
  - Several Scrum events, like the daily Scrum, would help the team communicate information in a timely manner. The sprint review would provide an opportunity for various stakeholders to review the work completed to date. The sprint retrospective would give the team time to reflect on and improve their work process.
- However, practice does not always follow theory, as described in the next section.

TEAM MEMBERS DO NOT HAVE ALL THE NECESSARY SKILLS

- Recall that one of the main reasons for using an agile approach is because you cannot define the scope of work upfront.
- If Scrum teams do not have any people on them who understand these technologies, how can they help develop policies to hire, retain, and educate workers with these skills? One approach would be to identify this lack of expertise as an impediment and raise it to senior management for resolution. They might decide to remove that work from the product backlog and work on finding a better solution. That solution might involve not just using agile.
- Example : the GCHC project was not a purely agile project.
  - Although they used Scrum, it was a modified version. They also agreed at the beginning of the project that they would follow whatever practices made sense. For example, to address the problem of not having internal expertise in areas like 3-D printing, robotics, and artificial intelligence, PM worked with other stakeholders to quickly develop and outsource several user stories related to those areas. One of those user stories involved researching the option to acquire other companies with those capabilities. Sometimes it makes sense to hire experts outside of the company (or acquire other companies) to meet specific needs. PM gladly handed over the management of that work to someone in another department.

TEAMS ARE NOT FOLLOWING SCRUM VALUES

- Effective teamwork is crucial to all projects, especially when using self-directed Scrum teams.
- Even though people are trained in agile and have worked on Scrum teams in the past, they still might have problems working together.
- Scrum theory emphasizes the values of commitment, focus, openness, respect, and courage, but these values are not easy to implement. What can you do if you see that teams are not working well together?
- It's important to address the root cause of the problem, especially with team dynamics.
  - For example, Debra (PM) noticed in the daily Scrum meetings with the incentives team that one team member, Scott, was very quiet, even after she asked him if he had anything to add. She knew Scott was introverted, but she also knew that he often had brilliant, innovative ideas. She pulled Scott aside after a meeting and openly discussed her concerns. It turned out the issue was that Scott was afraid to bring up ideas that differed from Vanessa's, who was a high-level manager and very aggressive. He also totally disagreed with some of her suggestions related to retention policies. Scott had been working virtually since the pandemic started, and if he was forced to go back into the office, he was ready to find another job. He knew many other workers felt the same way. Scott had trouble implementing the Scrum values of openness and courage. Vanessa was also Debra's boss, so she definitely related to this problem, knowing how intimidating Vanessa could be. Debra remembered a study on team effectiveness that said the number one quality for a great team was psychological safety. People have to feel safe to voice their opinions, especially if they differ from others. Debra talked to Vanessa and held a special meeting of the entire team to discuss how they would promote a safe culture. Vanessa acknowledged that she often came across pretty strong, but that she respected everyone's ideas and would be more aware of and open to seeking other opinions. She was very happy that Scott opened up about the issue of virtual work and rethought her ideas on policies related to it.

REQUIREMENTS CHANGE TOO QUICKLY

- Part of the beauty of an agile culture is the ability to pivot quickly.
- Product owners are responsible for prioritizing the product backlog before each sprint. But what if senior management keeps changing priorities, even after a sprint has started? What if the organization's culture is not receptive to such rapid changes?
- Sprint teams used to think that the sprint backlog could not change during a sprint.
- However, *The Scrum Guide* (2020) states the following: "The Sprint Goal is created during the Sprint Planning event and then added to the Sprint Backlog. As the Developers work during the Sprint, they keep the Sprint Goal in mind. If the work turns out to be different than they expected, they collaborate with the Product Owner to negotiate the scope of the Sprint Backlog within the Sprint without affecting the Sprint Goal."
- Therefore, agile teams have to be flexible and respond to changes very quickly.
- Leaders must empower teams to be change agents, always focused on adding value to the organization.
  - For example, Debra and her incentives team held a special meeting to address the competitor's new tuition policy. They might discover that the upfront reimbursement only applied to a few online programs and not most accredited colleges and universities. They could investigate ways their company could provide tuition assistance upfront in particular cases. They could also provide further incentives for employees to successfully complete courses. Her team could then clarify their new policies to address these issues.

CHAPTER SUMMARY

- Good execution is crucial to project success. Without it, the products, services, and results planned from the project cannot materialize.
- This chapter summarizes the executing processes and key outputs for project integration, quality, resource, communications,
stakeholder, risk, and procurement management. It also discusses important concepts and tools and techniques to help improve project
execution.
- Common outputs to these knowledge areas include change requests and updates to the project management plan, project documents, and organizational process assets.
  - Integration management: deliverables, work performance data, issue log, and lessons learned register
  - Quality management: Quality report, test and evaluation
  - Resource management: Physical resource assignments, project team assignments, resource calendars, and team performance assessments
  - Communications management: Project communications Procurement management: selected sellers and agreements
- It is also important to use effective leadership and apply several project management principles that assist in executing projects. Project managers must understand the theory as well as the practice of executing agile projects.